Handwritten Digit Classification using Neural Networks

Abstract

This project focuses on recognizing handwritten digits using Neural Networks. The system is trained using the MNIST dataset, which contains thousands of labeled images of digits from 0 to 9. The model learns patterns from the dataset and classifies handwritten digits with high accuracy. This project demonstrates the effectiveness of machine learning in image classification tasks.

Introduction

Handwritten digit classification is a significant application of machine learning and computer vision. It is widely used in real-world applications such as bank cheque processing, postal code recognition, and document digitization. Neural Networks are highly effective in identifying patterns in images, making them suitable for this task.

Objective

- To classify handwritten digits from 0 to 9
- To build a Neural Network model
- To achieve high prediction accuracy
- To understand image processing and classification

Dataset

The MNIST dataset is used in this project.

- Total images: 70,000
- Training data: 60,000 images
- Testing data: 10,000 images
- Image size: 28 × 28 pixels
- Type: Grayscale images

Methodology

1. Load the dataset
2. Preprocess the images (normalization)
3. Split into training and testing data
4. Build Neural Network model

5. Train the model
6. Evaluate performance
7. Predict handwritten digits

Model Used

- Artificial Neural Network (ANN)
- Activation Functions: ReLU, Softmax
- Optimizer: Adam
- Loss Function: Categorical Crossentropy

Results

The model achieves an accuracy of approximately **97% to 99%** on test data.
It successfully classifies most handwritten digits with high precision.

Applications

- Bank cheque reading
- Postal code recognition
- Digitizing handwritten forms
- Automated data entry systems

Conclusion

This project successfully demonstrates handwritten digit classification using Neural Networks. The model performs efficiently and achieves high accuracy, proving the effectiveness of deep learning techniques in image classification tasks.

Future Scope

- Improve accuracy using Convolutional Neural Networks (CNN)
- Develop a real-time application
- Extend to handwritten character recognition